MSG No. 101.179 $P4_2cm$ [Type I, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `2.mm`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,4,7,8]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[2,3,5,6]`

Table 2: Wyckoff site: `2b`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[1,4,7,8]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,3,5,6]`

Table 3: Wyckoff site: `4c`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\, \frac{1}{2},\, z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{2},\, 0,\, z\right]$	`[7,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `..m`

No.	position	mapping
1	$[x,\, x,\, z]$	`[1,8]`
2	$\left[-x,\, x,\, z+\frac{1}{2}\right]$	`[2,5]`
3	$\left[x,\, -x,\, z+\frac{1}{2}\right]$	`[3,6]`
4	$[-x,\, -x,\, z]$	`[4,7]`

Table 5: Wyckoff site: `8e`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	`[5]`
6	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	`[6]`
7	$[-y,\, -x,\, z]$	`[7]`
8	$[y,\, x,\, z]$	`[8]`